MISSING DATA & STATISTICAL INFERENCE

A Monte Carlo Study of MCAR, MAR and MNAR

Eight Ways of Handling Missing Values, and the Coverage of Classical, Bootstrap and Rubin Confidence Intervals

Project Report
M.Sc. Statistics
November 2026

Summary

Real datasets very often have missing values. This project asks: **if some values are missing, does the way we deal with them change our results, and can we still trust the confidence interval that comes with them?**

We created data from a model we know, $Y = 5 + 2X_1 - 1.5X_2 + X_3 + \text{error}$, with 200 observations, so the true slope is $\beta_1 = 2$. About 20% of the values of Y and of X_1 were deleted in three ways: completely at random (MCAR), depending on another observed variable (MAR), and depending on the value that disappears (MNAR). Eight methods were applied to every incomplete dataset: complete-case analysis, mean imputation, hot-deck, regression imputation, stochastic regression, two versions of multiple imputation (MICE) and the EM algorithm.

For every method we built the confidence intervals an analyst would actually use: the **classical** interval (from the usual formula), two **bootstrap** intervals (standard and percentile) and, for MICE, the **Rubin** interval. To find out whether these intervals can be trusted we ran a **Monte Carlo study**: 300 new datasets were drawn from the true model, and inside each one every method was bootstrapped 200 times (N × B replication). The **coverage** of an interval is the share of the 300 intervals that contain the true value; it should be about 95%. All comparisons are made **within** each mechanism.

Main findings

- **Mean imputation is the worst method in every setting.** Its bias is -0.40, -0.49 and -0.57 (MCAR, MAR, MNAR), and no interval — classical or bootstrap — contains the truth in more than 17.3% of the runs.
- **The classical interval of single imputation is too narrow.** For hot-deck, regression and stochastic regression the formula SE is only 68–79% of the true spread, because made-up values are treated as real. Stochastic regression is unbiased, yet its classical interval covers only 85% (MCAR) and 82% (MAR).
- **The bootstrap fixes the standard error, but not bias.** Because it repeats the imputation in every resample, the bootstrap SE is 92–99% of the true spread for *every* method. Stochastic regression with a bootstrap interval reaches 95% (MCAR) and 95% (MAR). But for biased methods the interval is centred in the wrong place, and the remaining under-coverage is predicted almost exactly by the bias.
- **MCAR:** complete case works well (classical coverage 95%); MICE and EM are slightly more precise (RMSE 0.087 and 0.084 against 0.091).
- **MAR:** complete case is biased (-0.103) and every one of its intervals covers below 80% — the bootstrap cannot help. EM and MICE (normal) are essentially unbiased with 93% and 94% coverage, because they use the extra observed variable.
- **MNAR:** no method and no interval reaches 95%; the best reach about 91%.

Contents

1 Introduction **3**

2 Aim and Research Question **3**

3 How the Data Were Created **3**

4 How Values Were Deleted **4**

5 The Eight Methods **5**

6 Confidence Intervals and How They Are Checked **6**

6.1 Four kinds of interval 6

6.2 The bootstrap 7

6.3 Why Monte Carlo is needed: bootstrap and Monte Carlo are different 7

6.4 What is measured 8

7 One Dataset: An Illustration **8**

8 Results of the Monte Carlo Study **9**

8.1 MCAR: the gaps are pure luck 9

8.2 MAR: the gaps depend on an observed variable 11

8.3 MNAR: the gaps depend on the missing value itself 13

8.4 Classical versus bootstrap intervals, one by one 15

8.5 Before and after the bootstrap 16

9 Discussion **17**

9.1 Mean imputation hurts everywhere 17

9.2 Why the classical interval of single imputation is too narrow 17

9.3 What the bootstrap fixes, and what it cannot 17

9.4 Bootstrap standard versus percentile 18

9.5 Multiple imputation versus single imputation plus bootstrap 18

9.6 PMM versus the normal version of MICE 18

9.7 MNAR is different in kind 18

10 Limitations **19**

11 Conclusion **19**

Appendix A: A Worked MICE Example **20**

Appendix B: Settings and How to Reproduce **20**

References **20**

1 Introduction

Missing data are very common: a survey respondent skips a question, a machine fails to record a reading, a patient leaves a study early. When values are missing, the analysis changes for two reasons. There is less information, and the way we *handle* the missing values can change the relationships we find and how sure we are about them.

How serious the problem is depends on *why* the values are missing. Statisticians describe three situations:

- **MCAR (Missing Completely At Random).** Whether a value is missing has nothing to do with the data. Example: a sensor fails at random moments.
- **MAR (Missing At Random).** Whether a value is missing depends on something we *did* observe. Example: older people skip a question more often, and age is recorded for everyone.
- **MNAR (Missing Not At Random).** Whether a value is missing depends on the missing value itself. Example: people with very high incomes refuse to report their income.

These form a ladder: MCAR ⊂ MAR ⊂ MNAR. There are many ways to deal with missing values, and they can lead to different estimates *and* to different confidence intervals. This project compares them in a controlled experiment where the true answer is known, and checks not only the estimates but also whether the confidence intervals really contain the truth as often as they claim.

2 Aim and Research Question

The aim is to find out how different missing-data methods affect a regression estimate and its uncertainty, when every method is given the same incomplete data from the same missingness mechanism. We focus on β_1, the slope of X_1, whose true value is 2.

> **The question.** *Within each type of missingness, which way of handling the missing values gives the most accurate estimate, and which confidence intervals (classical, bootstrap or Rubin) contain the true value about 95% of the time, as a 95% interval should?*

The question is answered three times: once for MCAR, once for MAR and once for MNAR. The three mechanisms are not ranked against each other; each is a separate setting in which the methods are compared.

3 How the Data Were Created

Each complete dataset follows the linear model

$$Y = 5 + 2X_1 - 1.5X_2 + 1X_3 + \varepsilon,$$

where X_1, X_2, X_3 and the error ε are independent standard normal variables, with **n = 200** observations. We also create an **auxiliary variable** $Z = Y$ + noise (correlation with Y about 0.94). Z is not in the regression and is never missing, but it can help to fill in the gaps, like a cheap rough measurement next to an exact one.

Parameter	True value	Estimate on the walkthrough dataset (nothing missing)	Standard error
β0 (intercept)	5.000	4.9629	0.0726
β1	2.000	2.0368	0.0722
β2	-1.500	-1.5076	0.0749
β3	1.000	0.9483	0.0776

Table 1: The regression on one complete dataset (seed 42), before any value is deleted.

The estimate of β_1 is 2.037, not exactly 2: a dataset is a sample, so even with nothing missing the estimate has random error. This is why one dataset can never tell us whether a method works.

4 How Values Were Deleted

About **20% of the values of Y and of X_1** were deleted, each separately, so roughly a third of the rows have at least one gap. The mechanisms differ only in *how likely* a value is to be deleted (Figure 1):

- **MCAR:** every value has the same 20% chance.
- **MAR:** the chance depends on Z, which is always observed; rows with a large Z lose values more often.
- **MNAR:** the chance depends on the value being deleted; large Y and large X_1 vanish more often.

For MAR and MNAR the chance is expit$(-2.23 + 2 \times d)$, where d is the standardised driver and expit$(z) = 1/(1 + e^{-z})$ turns any number into a probability. The -2.23 keeps the overall rate near 20% in every mechanism.

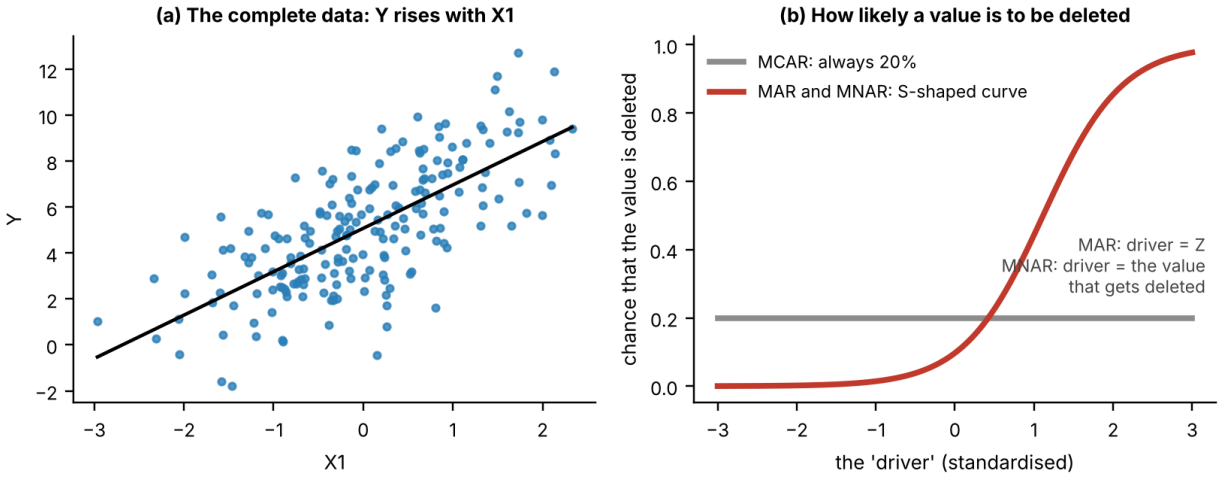


Figure 1: (a) The complete data. (b) The chance that a value is deleted under each mechanism.

Mechanism	X1 missing	Y missing	Rows with at least one gap	Complete rows left
MCAR	45	43	78 (39%)	122
MAR	39	40	61 (30%)	139
MNAR	41	41	65 (32%)	135

Table 2: Missing values in the walkthrough dataset (out of 200 rows).

Mechanism	Y missing (%)	X1 missing (%)	Rows with a gap (%)	Z: missing − observed	Y: missing − observed
MCAR	20.11	20.02	36.10	-0.01	-0.03
MAR	19.87	20.16	30.58	3.96	3.53
MNAR	19.82	20.08	32.28	3.74	3.72

Table 3: Check of the deletion rules on 20,000 rows. Under MCAR the rows that lose Y look like the others; under MAR and MNAR they have clearly higher Z and Y.

In simple words. In real data we can see Z but not the deleted Y values, so MAR and MNAR cannot be told apart from the observed data alone. We can tell them apart here only because we deleted the values ourselves.

5 The Eight Methods

Every method ends with the same regression of Y on X_1, X_2, X_3. They differ only in what they do with the gaps first.

Method	How it works	Good	Not so good
Complete case	Delete every row with a gap.	Simple. Fine under MCAR.	Throws away rows; can be biased otherwise.
Mean imputation	Fill each gap with the column average.	Very simple.	Shrinks the spread and weakens relationships.
Hot-deck	Copy the value from the most similar row.	Uses real values.	One guess per gap.
Regression imputation	Fill each gap with its regression prediction.	Uses the other variables.	Filled values sit exactly on the line.
Stochastic regression	Prediction plus a random residual.	Restores natural scatter.	Still one guess per gap.
MICE (normal)	Make 10 random completed datasets, analyse each, combine.	Honest uncertainty.	More work.
MICE (PMM)	Same, but borrow real values from similar rows.	Real values; honest uncertainty.	Needs enough similar donors.
EM	Maximum likelihood using every row, even partly observed ones.	Uses all information.	Assumes normality.

Table 4: The eight methods at a glance.

Regression imputation, stochastic regression and both MICE versions share one engine: each incomplete variable is regressed on all the others (Z included), and the variables are filled in one at a time, cycling through them five times. Regression imputation uses the prediction; stochastic regression adds a random residual; MICE first

draws new regression coefficients (so the uncertainty of the model itself is included), fills the gaps either from the regression model (normal) or by borrowing the real value of one of the five rows with the closest prediction (PMM), and repeats this 10 times. **EM** estimates the means and covariances of all five variables by maximum likelihood, alternating an expectation step (fill each gap with its expected value) and a maximization step (re-estimate), and computes the regression coefficients from them.

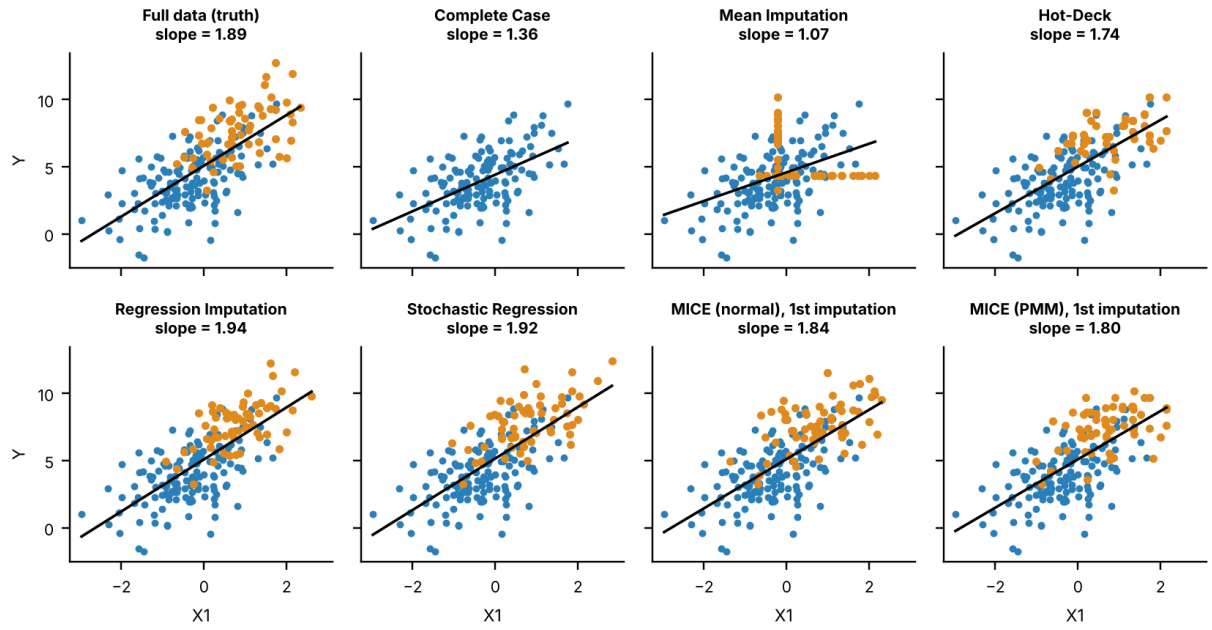


Figure 2: Y against X1 in the MAR walkthrough dataset after each method has handled the gaps. Orange = rows where a value was missing; the black line is the fitted slope of Y on X1.

Figure 2 shows what the methods do. Complete case loses the top-right part of the cloud (under MAR, rows with a large Z lose values), so its slope flattens to 1.36. Mean imputation puts the filled points on a cross through the average, flattening the slope to 1.07. Regression imputation places them exactly on the line. The other methods scatter them around the line like real data. (These slopes are of Y on X_1 alone; for the full data it is 1.89.)

6 Confidence Intervals and How They Are Checked

6.1 Four kinds of interval

Interval	How it is built	Used for
Classical	estimate ± 1.96 × the usual regression formula SE	complete case, mean, hot-deck, regression, stochastic
Bootstrap standard	estimate ± 1.96 × the bootstrap SE	complete case, mean, hot-deck, regression, stochastic, EM
Bootstrap percentile	the 2.5% and 97.5% points of the bootstrap estimates	the same six methods
Rubin	pooled estimate ± 1.96 × the Rubin's-rules SE	MICE (normal) and MICE (PMM)

Table 5: The confidence intervals compared.

For the imputation methods the classical interval is *naive*: it treats the filled-in values as if they were real observations. EM has no simple standard-error formula, so it gets only the bootstrap intervals. MICE needs no bootstrap, because Rubin's rules (total variance = within + (1 + 1/m) × between) already include the extra uncertainty caused by the missing data.

6.2 The bootstrap

The bootstrap does not know the true population, so it treats the sample as if it were the population: draw 200 rows *with replacement*, **redo the whole method** (including filling in the gaps), and record β_1. Repeating this 200 times gives 200 estimates. Their standard deviation is the **bootstrap SE** and their middle 95% is the **percentile interval**. Of the four common bootstrap intervals (standard, percentile, bootstrap-t and BCa), only the standard and bootstrap-t intervals use the bootstrap SE; we use the standard and percentile intervals.

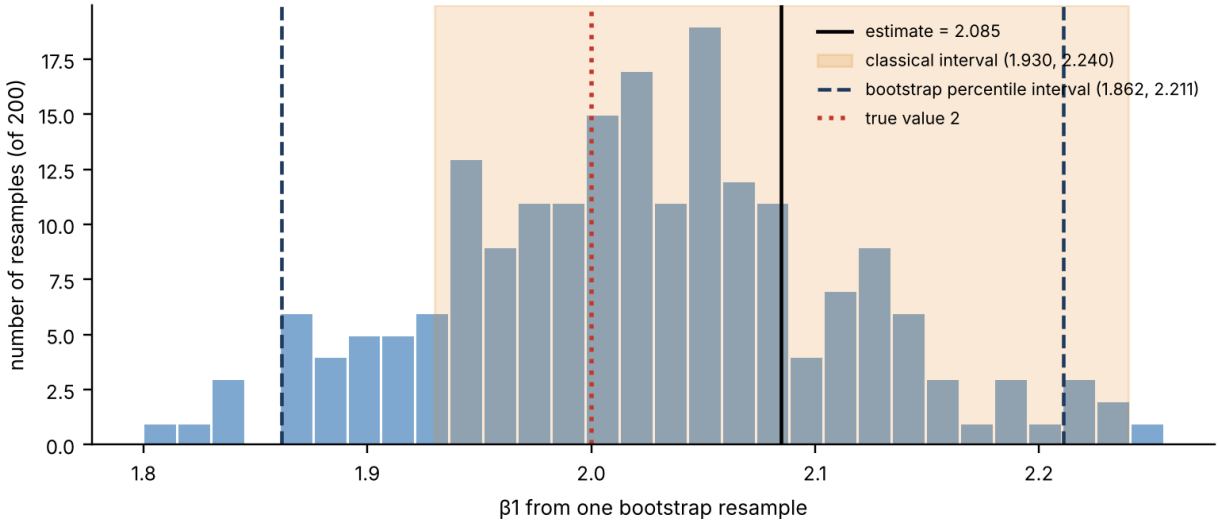


Figure 3: One bootstrap: 200 estimates of β1 for stochastic regression on the MAR walkthrough dataset.

Figure 3 shows one bootstrap. The classical interval of stochastic regression on this dataset is (1.930, 2.240); the percentile interval is (1.862, 2.211). Both happen to contain 2 here — but that is one dataset.

6.3 Why Monte Carlo is needed: bootstrap and Monte Carlo are different

- The **bootstrap** resamples the rows of *one* dataset and produces *one* confidence interval. It cannot say how often such intervals contain the truth.
- **Monte Carlo** draws *many new datasets from the true population*. Building the interval on each and counting how many contain the true value gives the **empirical coverage probability**. Without Monte Carlo the coverage of an interval cannot be found.

So the bootstrap runs **inside** the Monte Carlo. For each of 300 repetitions: (1) generate a new complete dataset from the true model; (2) for each mechanism, delete values with the same deletion function; (3) apply all eight methods, and for each build its classical

interval, bootstrap it 200 times for the two bootstrap intervals, and build the Rubin interval for MICE; (4) also analyse the complete data as a benchmark. This is **N × B replication**: about one million imputations and regressions in total.

6.4 What is measured

Measure	What it tells us
Bias	Average of (estimate − 2) over the 300 datasets. Zero means right on average.
Spread (SD)	How much the estimate varies from one dataset to the next: the true uncertainty.
RMSE	Overall accuracy, combining bias and spread. Smaller is better.
Coverage (%)	Share of the 300 intervals that contain 2. It should be about 95.
Width	Average length of the interval. Narrower is better, but only if coverage is right.
SE ÷ SD	Reported standard error divided by the true spread. 1 = honest; below 1 = over-confident.

Table 6: Measures used in the Monte Carlo study.

With 300 repetitions, a coverage near 95% has a standard error of about 1.3 percentage points, so the 95% band of random error is about ±2.5 points. Differences smaller than that should not be over-read.

7 One Dataset: An Illustration

Before the Monte Carlo study, every method was applied to one dataset (seed 42) under each mechanism, step by step, to see what it does. Table 7 shows the estimates. The complete-data estimate is 2.037.

Method	MCAR	MAR	MNAR
Complete Case	2.098	1.844	2.015
Mean Imputation	1.622	1.346	1.263
Hot-Deck	2.005	1.878	1.935
Regression Imputation	2.168	2.110	2.212
Stochastic Regression	2.060	2.085	2.120
MICE (normal)	2.082	2.036	2.107
MICE (PMM)	2.062	1.943	2.047
EM	2.076	2.022	2.108

Table 7: The β1 estimate of every method on the one walkthrough dataset.

On this one dataset mean imputation is far below the truth under every mechanism (1.62, 1.35, 1.26), and the other methods land between 1.84 and 2.21. One dataset is one random draw, so it cannot show which method is reliable on average or whether its interval can be trusted; that is what the Monte Carlo study is for.

8 Results of the Monte Carlo Study

For each mechanism we show a table of the estimates (ranked by RMSE), a table of the coverage of every interval, and a figure with three panels: (a) the spread of the 300 estimates, (b) the coverage of each interval type, and (c) the reported SE divided by the true spread. Colours in panel (a): grey = complete case, orange = single imputation, blue = MICE and EM; light grey = the benchmark with no missing data.

8.1 MCAR: the gaps are pure luck

Rank	Method	Bias	Spread (SD)	RMSE
–	Full data (benchmark)	-0.000	0.071	0.071
1	EM	-0.003	0.084	0.084
2	MICE (normal)	-0.006	0.087	0.087
3	MICE (PMM)	-0.011	0.087	0.087
4	Complete Case	-0.005	0.091	0.091
5	Stochastic Regression	-0.002	0.095	0.095
6	Regression Imputation	+0.081	0.084	0.117
7	Hot-Deck	-0.096	0.108	0.144
8	Mean Imputation	-0.397	0.141	0.421

Table 8: MCAR, the estimates (300 repetitions), ranked by RMSE.

Method	Classical	Bootstrap standard	Bootstrap percentile	Rubin	Width of best-suited interval
Full data (benchmark)	94.3	93.7	93.7	–	0.281
Complete Case	95.0	93.7	92.3	–	0.344
Mean Imputation	14.7	17.3	13.7	–	0.527
Hot-Deck	73.0	85.0	84.3	–	0.407
Regression Imputation	68.7	83.0	78.7	–	0.319
Stochastic Regression	85.0	92.0	95.0	–	0.342
MICE (normal)	–	–	–	95.3	0.329
MICE (PMM)	–	–	–	94.0	0.327
EM	–	93.3	92.7	–	0.317

Table 9: MCAR, coverage (%) of every 95% interval (target 95; random error about ±2.5). Width: percentile interval (Rubin for MICE, classical for the benchmark).

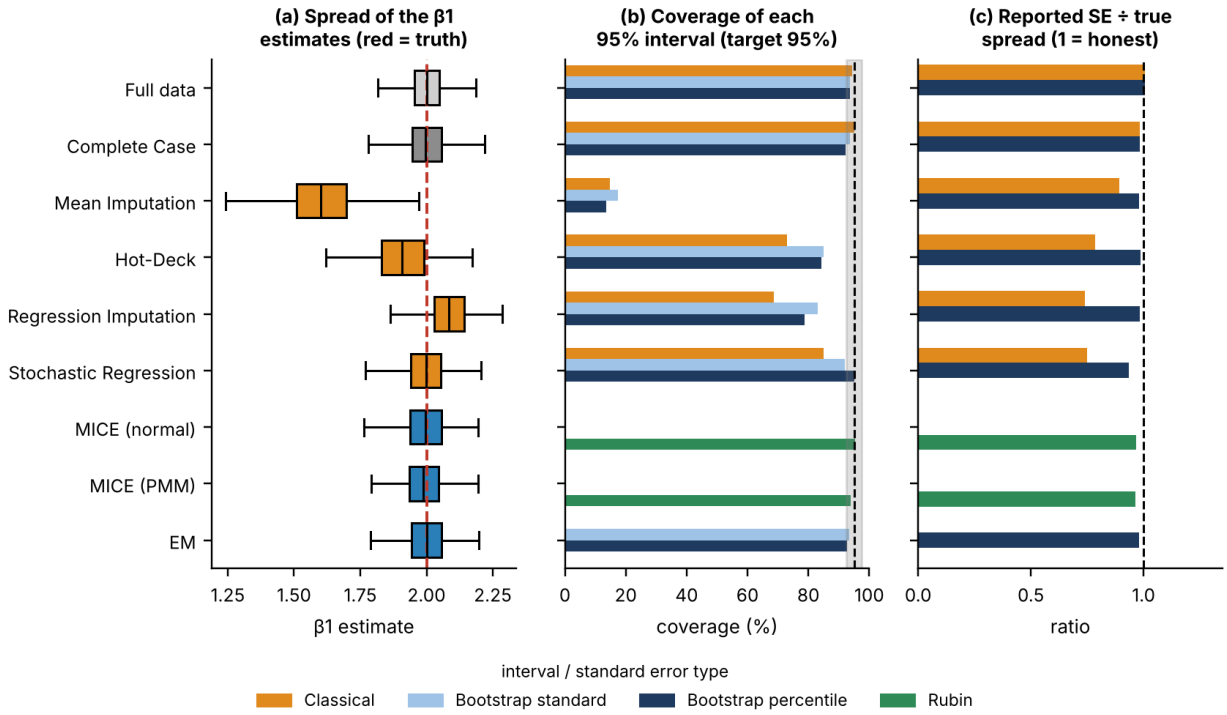


Figure 4: MCAR. (a) Spread of the estimates; (b) coverage of each interval type; (c) reported SE ÷ true spread.

- **Complete case is fine.** Unbiased (-0.005) with coverage 95% (classical) and 92% (bootstrap percentile). The rows that remain are a fair sample; the only cost is precision.
- **MICE and EM** are unbiased and slightly more accurate (RMSE 0.087, 0.087, 0.084), with coverage 95%, 94% (Rubin) and 93% (EM, bootstrap).
- **Stochastic regression** is unbiased (-0.002). Its classical interval covers only 85%, but with the bootstrap it reaches 95% (percentile).
- **Hot-deck and regression imputation** are biased (-0.096 and +0.081). The bootstrap raises their coverage (hot-deck 73% → 85%, regression 69% → 83%) but not to 95%. Mean imputation reaches at most 17.3%.

8.2 MAR: the gaps depend on an observed variable

Rank	Method	Bias	Spread (SD)	RMSE
–	Full data (benchmark)	-0.000	0.071	0.071
1	EM	+0.001	0.090	0.090
2	MICE (normal)	-0.003	0.090	0.090
3	Stochastic Regression	+0.003	0.098	0.098
4	MICE (PMM)	-0.050	0.100	0.112
5	Regression Imputation	+0.082	0.091	0.123
6	Complete Case	-0.103	0.100	0.144
7	Hot-Deck	-0.151	0.124	0.196
8	Mean Imputation	-0.485	0.140	0.505

Table 10: MAR, the estimates (300 repetitions), ranked by RMSE.

Method	Classical	Bootstrap standard	Bootstrap percentile	Rubin	Width of best-suited interval
Full data (benchmark)	94.3	93.7	93.7	–	0.281
Complete Case	77.0	75.0	73.3	–	0.355
Mean Imputation	4.7	4.3	3.7	–	0.500
Hot-Deck	57.3	73.7	72.3	–	0.459
Regression Imputation	66.7	84.7	81.3	–	0.340
Stochastic Regression	82.3	94.7	94.7	–	0.359
MICE (normal)	–	–	–	94.0	0.343
MICE (PMM)	–	–	–	87.0	0.355
EM	–	93.0	92.7	–	0.332

Table 11: MAR, coverage (%) of every 95% interval (target 95; random error about ±2.5). Width: percentile interval (Rubin for MICE, classical for the benchmark).

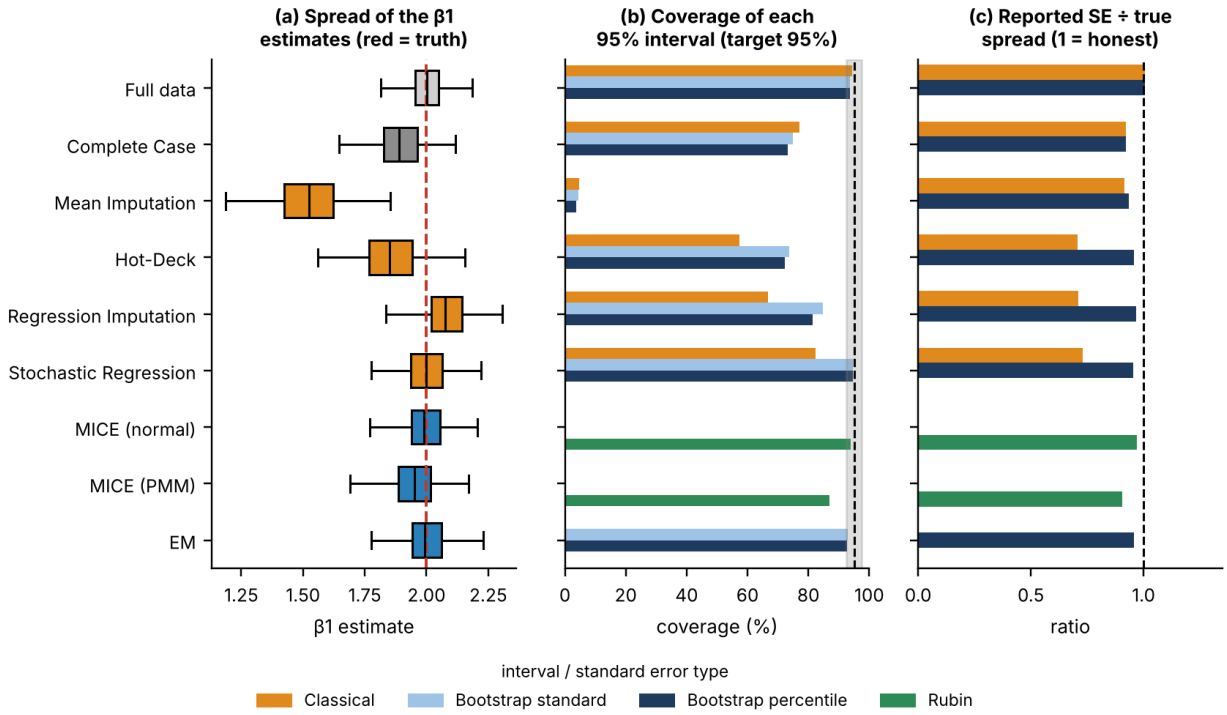


Figure 5: MAR. (a) Spread of the estimates; (b) coverage of each interval type; (c) reported SE ÷ true spread.

- **Complete case is no longer safe.** It is biased (-0.103) because the rows that lose values have large Z, which goes with large Y. Its coverage is 77% (classical), 75% and 73% (bootstrap): its standard error is fine, the problem is where the interval is centred.
- **EM and MICE (normal)** use Z, are essentially unbiased (+0.001, -0.003), have the lowest RMSE (0.090, 0.090) and coverage 93% (EM, bootstrap) and 94% (Rubin).
- **Stochastic regression with the bootstrap** also works: unbiased (+0.003), coverage 82% with the classical interval but 95% with the bootstrap percentile interval.
- **MICE (PMM)** has a small downward bias (-0.050) and 87% coverage. Hot-deck (72%) and regression imputation (81%) improve with the bootstrap but stay biased; mean imputation stays near 4%.

8.3 MNAR: the gaps depend on the missing value itself

Rank	Method	Bias	Spread (SD)	RMSE
–	Full data (benchmark)	-0.000	0.071	0.071
1	MICE (PMM)	+0.019	0.117	0.119
2	MICE (normal)	+0.068	0.099	0.120
3	EM	+0.071	0.099	0.122
4	Complete Case	-0.060	0.108	0.124
5	Stochastic Regression	+0.074	0.107	0.129
6	Hot-Deck	-0.047	0.127	0.135
7	Regression Imputation	+0.157	0.103	0.188
8	Mean Imputation	-0.573	0.158	0.594

Table 12: MNAR, the estimates (300 repetitions), ranked by RMSE.

Method	Classical	Bootstrap standard	Bootstrap percentile	Rubin	Width of best-suited interval
Full data (benchmark)	94.3	93.7	93.7	–	0.281
Complete Case	87.0	87.0	86.3	–	0.393
Mean Imputation	4.3	4.0	4.0	–	0.572
Hot-Deck	84.7	91.3	90.7	–	0.469
Regression Imputation	41.7	61.3	57.3	–	0.365
Stochastic Regression	75.0	87.3	88.3	–	0.385
MICE (normal)	–	–	–	89.0	0.372
MICE (PMM)	–	–	–	89.0	0.383
EM	–	87.0	84.3	–	0.354

Table 13: MNAR, coverage (%) of every 95% interval (target 95; random error about ±2.5). Width: percentile interval (Rubin for MICE, classical for the benchmark).

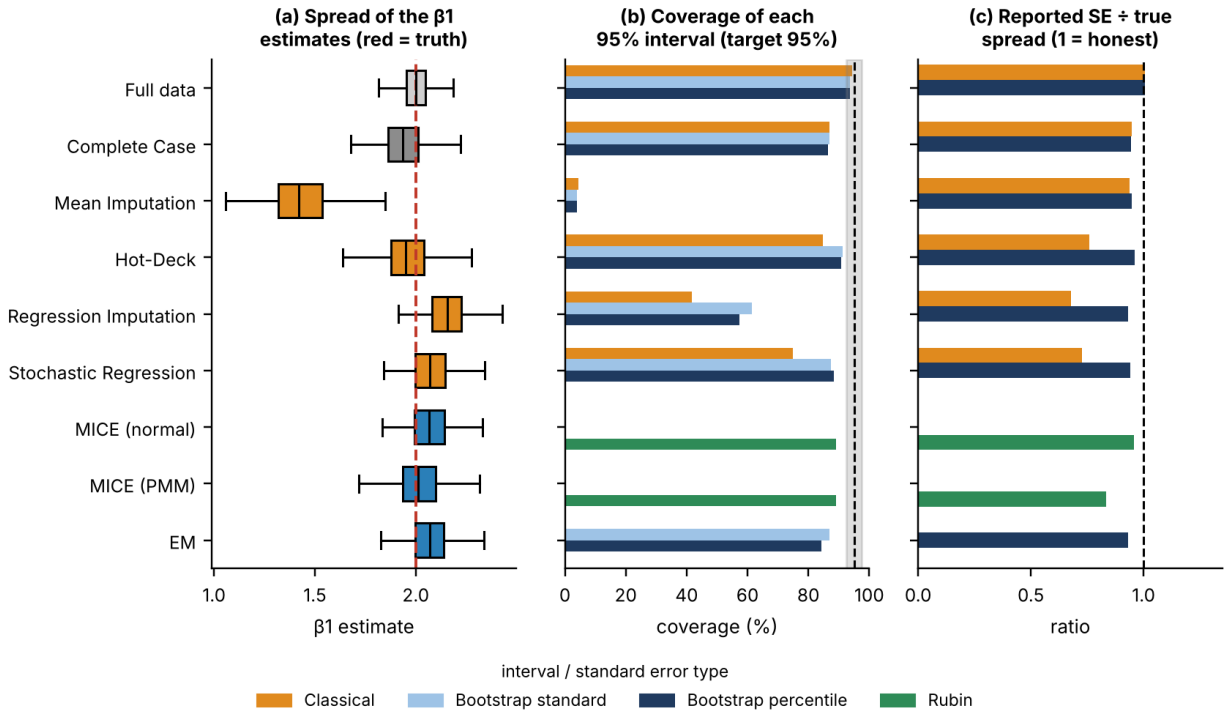


Figure 6: MNAR. (a) Spread of the estimates; (b) coverage of each interval type; (c) reported SE ÷ true spread.

- **No method is reliable.** Complete case is biased downward (-0.060); EM, MICE (normal) and stochastic regression, which all assume MAR, overshoot (+0.071, +0.068, +0.074). The highest coverage of any interval is 91%, below the 95% target even after allowing for random error.
- **The bootstrap helps a little but cannot remove MNAR bias.** Stochastic regression goes from 75% to 88%, regression imputation from 42% to 57%.
- Hot-deck happens to have a small bias here (-0.047) and reaches 91% with the bootstrap; this is most likely two errors cancelling, not a sign that hot-deck handles MNAR.

8.4 Classical versus bootstrap intervals, one by one

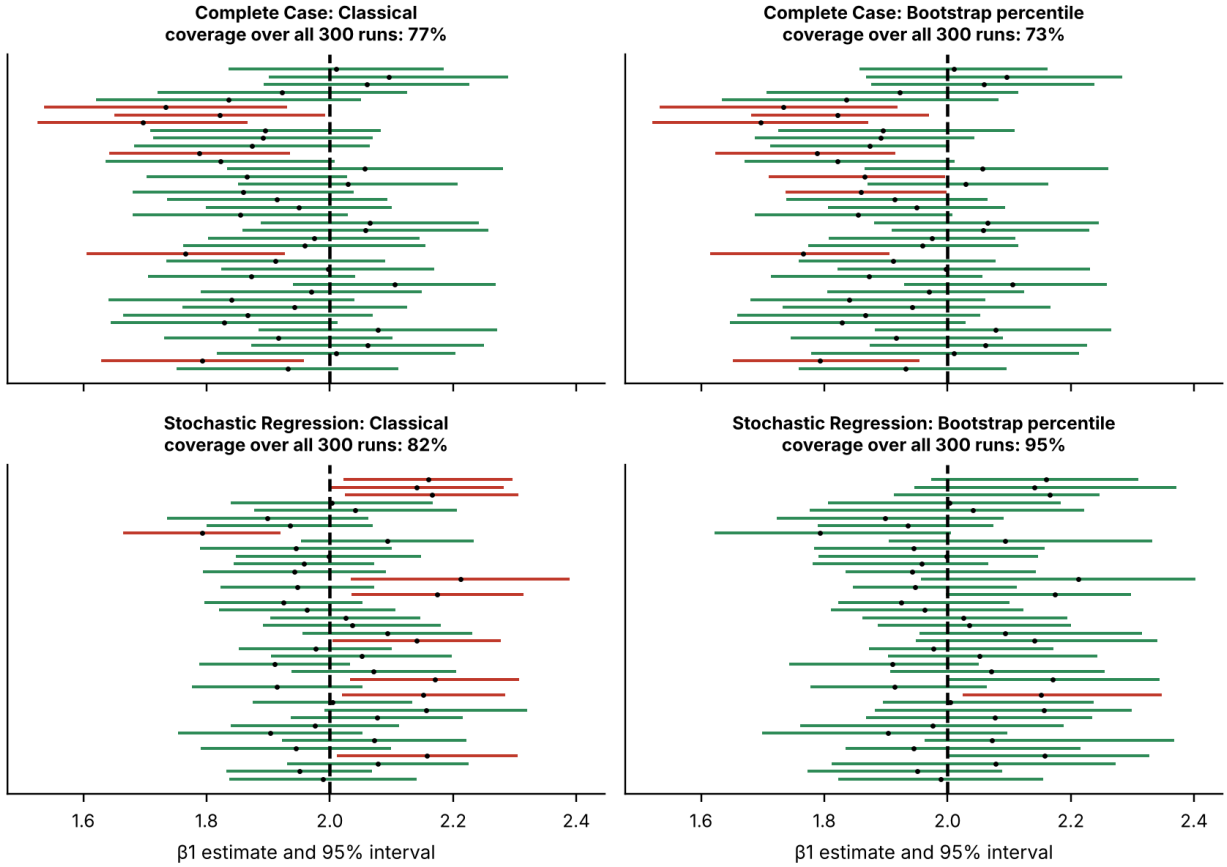


Figure 7: MAR. The first 40 intervals of complete case and stochastic regression: classical (left) and bootstrap percentile (right). Green = contains the truth, red = misses.

Figure 7 makes the difference visible. Complete case misses mostly on the *low* side whichever interval is used: the intervals are centred too low (bias), and making them honest does not move them. Stochastic regression's classical intervals are centred correctly but too short, so they miss on both sides; its bootstrap intervals are about 28% wider and the coverage rises from 82% to 95%.

8.5 Before and after the bootstrap

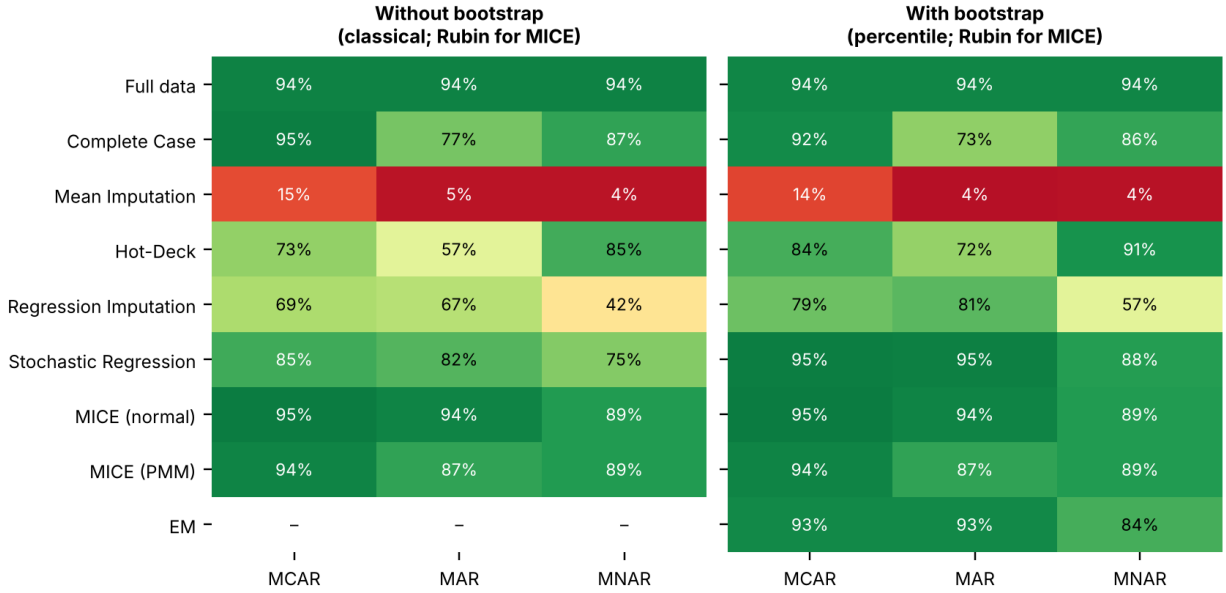


Figure 8: Coverage of every method under every mechanism, without the bootstrap (left) and with the bootstrap percentile interval (right). MICE uses the Rubin interval in both.

Figure 8 summarises everything. Moving from the left map (no bootstrap) to the right map (bootstrap percentile), the cells of stochastic regression turn dark green under MCAR and MAR, and hot-deck and regression imputation improve — the bootstrap repairs over-confidence. The cells of mean imputation stay red, complete case under MAR stays light, and the MNAR column has no dark green cell in either map — the bootstrap cannot repair bias. MICE (Rubin) and EM are dark green under MCAR and MAR in both maps.

Mechanism	Lowest RMSE	Smallest bias	Coverage closest to 95%*	Lowest RMSE among methods with coverage ≥ 90%*
MCAR	EM	Stochastic Regression	Stochastic Regression	EM
MAR	EM	EM	Stochastic Regression	EM
MNAR	MICE (PMM)	MICE (PMM)	Hot-Deck	Hot-Deck

*Table 14: The best method under each mechanism. *Coverage of the best-suited interval: Rubin for MICE, bootstrap percentile otherwise.*

Where the leading methods are within random error of each other, the “best” in this table should not be read as a real difference.

9 Discussion

9.1 Mean imputation hurts everywhere

Filling every gap with the same number removes the natural variation in Y and X_1, and with it part of their relationship, so the slope is pulled towards zero (Figure 2). The bias (-0.40 to -0.57) is several times the spread, so no interval — classical or bootstrap — can rescue it.

9.2 Why the classical interval of single imputation is too narrow

Hot-deck, regression and stochastic regression fill each gap once and then treat the completed data as real. About a third of the rows contain made-up values, so the formula thinks it has more information than it has: the classical SE is only 68–79% of the true spread (panel (c) of Figures 4–6), and the intervals are too short.

9.3 What the bootstrap fixes, and what it cannot

The bootstrap re-runs the *whole* method, filling-in included, on every resample. The spread of the 200 answers therefore contains the uncertainty created by the imputation, and the bootstrap SE is 92–99% of the true spread for every method. What the bootstrap cannot change is where the interval is centred: it resamples the *observed* data, so a biased method is equally biased in every resample. Table 16 shows that, once the SE is honest, the coverage of a method is almost exactly what its bias predicts: an interval centred on a normal estimate with bias b and spread SD covers the truth with probability $\Phi(1.96 \cdot r - b/SD) - \Phi(-1.96 \cdot r - b/SD)$, where r is the SE ÷ SD ratio. **The bootstrap repairs variance, not bias.**

Mechanism	Method	Bias ÷ SD	Bootstrap SE ÷ SD	Predicted coverage	Observed coverage
MCAR	Hot-Deck	-0.90	0.99	84.7%	85.0%
MCAR	Mean Imputation	-2.82	0.98	18.4%	17.3%
MAR	Complete Case	-1.03	0.92	77.8%	75.0%
MAR	Hot-Deck	-1.22	0.96	74.5%	73.7%
MAR	Regression Imputation	+0.90	0.97	83.7%	84.7%
MAR	Stochastic Regression	+0.03	0.95	93.8%	94.7%
MNAR	Regression Imputation	+1.53	0.93	61.3%	61.3%
MNAR	Stochastic Regression	+0.69	0.94	87.0%	87.3%

Table 16: Coverage of the bootstrap-standard interval predicted from bias and SD alone, and the coverage observed in the Monte Carlo study.

9.4 Bootstrap standard versus percentile

The two bootstrap intervals give very similar coverage here, because with n = 200 the bootstrap estimates are close to normal. With no missing data both cover 93.7%, slightly below 95% but within random error; bootstrap intervals are known to be a little too narrow in moderate samples, which is one reason refined versions (bootstrap-t, BCa) exist.

9.5 Multiple imputation versus single imputation plus bootstrap

Under MCAR and MAR, MICE (normal) with Rubin's rules and stochastic regression with a bootstrap interval reach the same goal — an unbiased estimate with an honest interval. MICE needs only 10 imputations; the bootstrap needs 200 complete re-runs of the method. Rubin's rules are the cheaper way to carry the imputation uncertainty; the bootstrap is the general-purpose way that works for any method, including EM.

9.6 PMM versus the normal version of MICE

Under MAR, MICE (PMM) is slightly biased (-0.050) with 87% coverage. The data are exactly linear and normal, so the normal model is right and PMM's robustness to odd distributions is not needed. PMM can only borrow observed values, and the rows that MAR targets (with extreme predicted values) have few similar donors.

9.7 MNAR is different in kind

Under MNAR the chance of being missing depends on the missing value itself, so the observed data cannot reveal how the missing values differ. Neither better imputation nor the bootstrap can fix this; the honest approach is a sensitivity analysis under several plausible assumptions.

10 Limitations

- **One simple model.** Linear, normal, independent predictors, n = 200, about 20% missing in two variables. Results can change with other distributions, sample sizes or amounts of missing data; in particular PMM is designed for non-normal data.
- **Chosen strengths and a strong helper variable.** The MAR and MNAR strengths were chosen by us, and Z is a close stand-in for Y. Real auxiliary variables are usually weaker.
- **Bootstrap choices.** 200 resamples per dataset is enough for the SE, but the 2.5% and 97.5% points rest on about 5 resamples each, so percentile limits are a little noisy. Bootstrap-t (which needs a second bootstrap inside every resample, N × B × B′ fits) and BCa intervals were not computed, and MICE was not bootstrapped (Rubin's rules were used instead).
- **Simplified MICE and EM.** MICE uses 5 cycles, 10 imputations, 5 donors and a normal (±1.96) interval rather than Rubin's t-based interval; EM assumes normality. For real work use a tested package such as *mice* in R or *statsmodels*.
- **Monte Carlo error.** With 300 repetitions, coverage near 95% is accurate to about ±2.5 points.
- **Only β_1 is examined**, not the other coefficients or predictions.

11 Conclusion

How missing data are handled is part of the statistical analysis, not a detail to be skipped. It changes the estimate and it changes whether the confidence interval can be trusted.

Mechanism	What works	What to avoid
MCAR	Complete case with its classical interval; MICE or EM for slightly more precision.	Mean imputation; classical intervals after single imputation.
MAR	EM (bootstrap interval) or MICE (normal, Rubin interval), with the helper variable in the imputation model; stochastic regression only with a bootstrap interval.	Complete case (biased); single imputation with classical intervals (over-confident).
MNAR	Nothing is reliable. Report a sensitivity analysis.	Claiming that any method or interval fixes MNAR.

Table 15: Recommendations by mechanism.

- **Never use mean imputation** for inference.
- **Never use the classical interval after single imputation**: it treats guesses as data. Use Rubin's rules, EM with a bootstrap, or a bootstrap that repeats the imputation.
- **The bootstrap fixes over-confidence, not bias.** An honest interval around a biased estimate still misses; choose a method that removes the bias first (under MAR: use the variables that explain the missingness).
- **Coverage can only be judged by Monte Carlo.** A single bootstrap interval says nothing about how often such intervals contain the truth.

Appendix A: A Worked MICE Example

The numbers behind MICE (normal) on the three walkthrough datasets: the 10 estimates of β_1 and Rubin's rules.

Mechanism	m1	m2	m3	m4	m5	m6	m7	m8	m9	m10
MCAR	2.097	2.089	2.101	2.083	2.035	2.076	2.162	2.087	2.031	2.055
MAR	1.948	2.014	2.012	2.073	2.093	2.081	2.102	2.048	2.055	1.930
MNAR	2.087	2.132	2.070	2.148	2.104	2.120	2.092	2.123	2.089	2.101

Table A1: The estimates of β1 from the completed datasets.

Mechanism	Pooled β1	Within (W)	Between (B)	Total = W + 1.1 B	Pooled SE
MCAR	2.082	0.00511	0.00140	0.00665	0.082
MAR	2.036	0.00583	0.00349	0.00967	0.098
MNAR	2.107	0.00616	0.00057	0.00679	0.082

Table A2: Rubin's rules.

Appendix B: Settings and How to Reproduce

Setting	Value
Observations per dataset	200
True model	Y = 5 + 2·X1 − 1.5·X2 + 1·X3 + ε, Z = Y + noise
Missing values	about 20% of Y and about 20% of X1
Walkthrough	one dataset, seed 42
Monte Carlo	300 repetitions, seed 2024 (one independent seed per repetition)
Bootstrap	200 resamples inside every dataset, for complete case, mean, hot-deck, regression, stochastic and EM
MICE	10 imputations, 5 cycles, PMM with 5 donors
Software	Python: numpy, pandas, scipy, scikit-learn, statsmodels, matplotlib, joblib (parallel computing)
Run time	the Monte Carlo study takes about 40 minutes on one core and is spread over all cores; results are saved and reloaded

References

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–38.

Efron, B., and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.

Rubin, D. B. (1987). *Multiple Imputation for Nonresponse in Surveys*. Wiley.

van Buuren, S. (2018). *Flexible Imputation of Missing Data*, 2nd ed. Chapman & Hall/CRC.

Koley, D. Missing Data – Unit I: Rigorous Treatment. Course material, Department of Statistics, St. Xavier’s University, Kolkata.

Koley, D. Missing Data – Unit II: Imputation Methods and Algorithmic Approaches. Course material, Department of Statistics, St. Xavier’s University, Kolkata.